Valid Adaptive Subgroup Discovery for Prediction-Powered Inference

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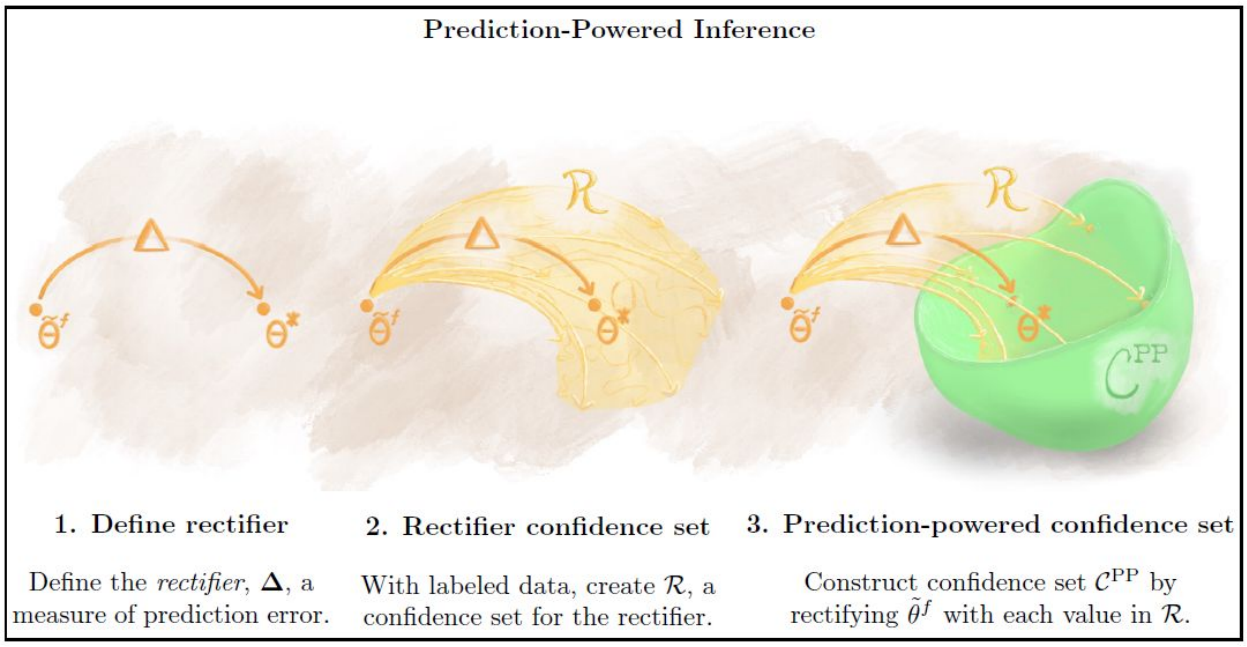
1 Introduction & Summary

In practice, researchers frequently have access to a small, "gold-standard" dataset. While it is possible to do statistical inference on this data, the small sample size severely limits the scope and magnitude of possible discoveries. Alternatively, researchers have also tried using a machine-learning system that can quickly generate predictions (i.e. unlabeled data), and treat these predictors as if they were "gold-standard" measurements. However, the resulting conclusions may be statistically invalid, as these predictions inevitably contain biases and errors.

Angelopoulos et al., 2023 introduces Prediction-Powered Inference (PPI), a statistical framework designed to perform valid inference by combining the smaller dataset with machine-learning predictors generated from a larger, unlabeled dataset of size N, where $N \gg n$. The core to the PPI framework is the "rectifier," denoted $\mathbf{\Delta}$. The rectifier captures exactly how errors in the machine-learning predictions lead to bias in the imputed estimate. To perform inference, the PPI framework first uses the smaller, gold-standard data to construct a confidence set for the rectifier. It then computes the imputed estimate on the massive unlabeled dataset and forms a final confidence set by adjusting the imputed estimate with every possible value in the rectifier's confidence set.



Critically, this method reduces the variance of the confidence set, since this framework takes into account the imputed data. In addition, PPI makes no assumptions regarding the machine-learning algorithm that generates the predictions. Finally, the PPI framework guarantees statistical validity in the familiar classical setting.

By utilizing a small labeled dataset to "rectify" predictions on a large unlabeled set, one can achieve narrower confidence intervals without assuming model accuracy. However, a major limitation remains: the choice of which subgroups to analyze is often determined post-hoc, leading to selection bias.

This project studies how to adapt prediction-powered inference to data-driven subgroup discovery. We fully develop and empirically validate a sample-splitting approach for adaptive PPI, extending it to a multi-region discovery pipeline with BH-based FDR control. In parallel, we derive a randomized-inference alternative and use theory to characterize regimes where it may outperform splitting; full empirical validation of that branch is left to future work.

Formalization of The PPI Framework The Prediction-Powered Inference (PPI) framework Angelopoulos et al., 2023 addresses the challenge of using machine-learning predictions $\hat{Y} = f(X)$ to perform valid statistical inference on a target quantity θ^* (e.g., the population mean μ_y). Given a small labeled dataset $\{(X_i, Y_i)\}_{i=1}^n$ and a large unlabeled dataset $\{X_j\}_{j=n+1}^{n+N}$, the PPI estimator for the mean is:

$$\hat{\theta}^{PPI} = \underbrace{\frac{1}{N}\sum_{j=n+1}^{n+N} f(X_j)}_{\text{Imputed Mean}} + \underbrace{\frac{1}{n}\sum_{i=1}^{n}(Y_i - f(X_i))}_{\text{Rectifier } \Delta} \tag{1}$$

The "rectifier" term Δ ensures that even if the model f is biased or poorly calibrated, the estimator remains unbiased for the true population mean. The resulting confidence intervals are valid asymptotically via the Central Limit Theorem.

2 Specifics, Proofs & Theorems

2.1 Problem Setup & Notation

Let $\{(X_i, Y_i)\}_{i=1}^n$ be a small labeled dataset drawn i.i.d. from acdistribution P, and let $\{\tilde{X}_j\}_{j=1}^N$ be a large unlabeled dataset also drawn i.i.d. from P, where $N \gg n$. A pretrained model f, independent of both datasets, produces predictions $\hat{Y}_i = f(X_i)$ on the labeled data and $\hat{\tilde{Y}}_j = f(\tilde{X}_j)$ on the unlabeled data.

The goal is valid inference on a population-level quantity

$$\theta^* = \arg\min_{\theta \in \mathbb{R}^p} \mathbb{E}\big[\ell_\theta(X, Y)\big],$$

for a loss function ℓ_θ that is convex in θ. Examples include means ($\ell_\theta(y) = \frac{1}{2}(y - \theta)^2$), quantiles (pinball loss), and linear/logistic regression coefficients.

2.2 Two Naive Approaches and Why They Fail

Classical approach (labeled data only). Using only the n labeled points, the confidence interval width scales as $\hat{\sigma}_Y/\sqrt{n}$. For small n, this produces wide intervals unlikely to yield significant results.

Imputation approach (predictions only). Imputation approach (predictions only). Using the N unlabeled predictions gives a narrow but potentially biased estimate: $\hat{\theta}^{\text{naive}}$ targets $\mathbb{E}[\ell_\theta(X, f(X))]$, not $\mathbb{E}[\ell_\theta(X, Y)]$, so the resulting confidence interval may fail to cover θ^*.

2.3 The PPI Estimator

PPI corrects the imputation bias. The rectifier $\Delta_\theta = \mathbb{E}[g_\theta(X, Y) - g_\theta(X, f(X))]$ measures the systematic gap between true and predicted subgradients. The PPI estimator minimizes the rectified loss:

$$\hat{\theta}^{\text{PP}} = \arg\min_\theta \left\{ \frac{1}{N}\sum_{j=1}^{N} \ell_\theta(\tilde{X}_j, f(\tilde{X}_j)) + \frac{1}{n}\sum_{i=1}^{n}\big[\ell_\theta(X_i, Y_i) - \ell_\theta(X_i, f(X_i))\big] \right\}.$$

The second sum is the empirical rectifier, which corrects the imputed risk so the estimator is unbiased for θ^*.

2.4 Constructing a Valid Confidence Set

The construction has two parts: a confidence set $\mathcal{R}_\delta(\theta)$ for the rectifier Δ_θ built from the n labeled points, and a confidence set $\mathcal{T}_{\alpha-\delta}(\theta)$ for the imputed gradient g_θ^f built from the N unlabeled points. Combining these via a Minkowski sum gives:

Combining. The key result (Theorem 1 of the paper, reproduced below) combines these two sets:

Theorem 2.1 (Convex Estimation; Angelopoulos et al., 2023)**.** *Fix $\alpha \in (0,1)$ and $\delta \in (0,\alpha)$. Suppose that for any $\theta \in \mathbb{R}^p$ we can construct $\mathcal{R}_\delta(\theta)$ and $\mathcal{T}_{\alpha-\delta}(\theta)$ as above. Let*

$$\mathcal{C}_\alpha^{\mathrm{PP}} = \big\{\theta : 0 \in \mathcal{R}_\delta(\theta) + \mathcal{T}_{\alpha-\delta}(\theta)\big\},$$

where $+$ denotes the Minkowski sum. Then $P(\theta^* \in \mathcal{C}_\alpha^{\mathrm{PP}}) \geq 1-\alpha$.

Proof sketch. By a union bound on the two confidence events, $0 = \Delta_{\theta^*} + g^f_{\theta^*} \in \mathcal{R}_\delta(\theta^*) + \mathcal{T}_{\alpha-\delta}(\theta^*)$, so $\theta^* \in \mathcal{C}_\alpha^{\mathrm{PP}}$. See Angelopoulos et al. (2023) for details.

2.5 Worked Example: Mean Estimation

Suppose the target statistic to estimate is

$$\theta^* = \mathbb{E}[Y_i].$$

The mean can also be expressed as the solution to a convex optimization problem:

$$\theta^* = \arg\min_{\theta\in\mathbb{R}} \mathbb{E}[\ell_\theta(Y_i)] = \arg\min_{\theta\in\mathbb{R}} \mathbb{E}\left[\frac{1}{2}(Y_i-\theta)^2\right].$$

In this case, the gradient is $g_\theta(y) = \theta - y$, so the rectifier is

$$\boldsymbol{\Delta} = \mathbb{E}[f(X_i) - Y_i].$$

Then, the suggested algorithm for prediction-powered mean estimation is as follows:

$$\begin{aligned}
\hat{\theta}^{\mathrm{PP}} &= \tilde{\theta}^f - \hat{\boldsymbol{\Delta}} \\
&= \frac{1}{N}\sum_{i=1}^{N} f(\tilde{X}_i) - \frac{1}{n}\sum_{i=1}^{n}(f(X_i) - Y_i) \\
\hat{\sigma}_f^2 &= \frac{1}{N}\sum_{i=1}^{n}(f(\tilde{X}_i - \tilde{\theta}^f)^2 && \text{empirical variance for } \mathcal{T} \\
\hat{\sigma}_{f-Y}^2 &= \frac{1}{n}\sum_{i=1}^{n}(f(X_i) - Y_i - \hat{\boldsymbol{\Delta}})^2 && \text{empirical variance for } \mathcal{R} \\
\omega_\alpha &= z_{1-\alpha/2}\sqrt{\frac{\hat{\sigma}_{f-Y}^2}{n} + \frac{\hat{\sigma}_f^2}{N}}
\end{aligned}$$

So, the reported confidence set is

$$\boxed{\mathcal{C}_\alpha^{\mathrm{PP}} = \left(\hat{\theta}^{\mathrm{PP}} \pm \omega_\alpha\right)}$$

2.5.1 Other Applications

PPI applies broadly to any estimand expressible as a solution to an estimating equation $\mathbb{E}[g(X,Y;\theta)] = 0$. Specific instances include:

Estimand	**Loss / Estimating Equation**
Mean	$\ell(X,Y;\theta) = (Y-\theta)^2$
Quantile	$\ell(X,Y;\theta) = \rho_\tau(Y-\theta)$
OLS coefficients	$g(X,Y;\theta) = X(Y - X^\top\theta)$
Logistic regression	$g(X,Y;\theta) = X(Y - \sigma(X^\top\theta))$

2.6 Prediction Powered p-values

The PPI framework can also be repurposed to compute valid prediction-powered p-values. For level α, the reported confidence interval was

$$C_\alpha^{\mathrm{PP}} = \left(\hat{\theta}^{\mathrm{PP}} \pm \omega_\alpha\right).$$

Therefore, the corresponding p-value is

$$p^{\mathrm{PP}} = \inf\left\{\alpha : \theta_0 \notin \left(\hat{\theta}^{\mathrm{PP}} \pm \omega_\alpha\right)\right\},$$

since p-values and confidence intervals are reciprocal methods for hypothesis testing. For a one-dimensional null hypothesis (ie $H_0 : \theta^* = \theta_0$), this can more simply be expressed as

$$p^{\mathrm{PP}} = \inf\left\{\alpha : |\hat{\theta}^{\mathrm{PP}} - \theta_0| > \omega_\alpha\right\}.$$

2.7 Assumptions & Limitations

1. **Power depends on model quality.** PPI narrows intervals only when $\sigma^2_{f-Y} < \sigma^2_Y$, i.e., the model explains some outcome variance. A model with $\sigma^2_{f-Y} > \sigma^2_Y$ makes PPI intervals wider than classical ones.
2. **Independence of f.** The prediction rule f must be pretrained on data independent of both the labeled and unlabeled datasets. Using the labeled data to train f destroys the rectifier's unbiasedness and invalidates coverage.
3. **Curse of dimensionality.** For non-linear estimands (quantiles, logistic regression), the confidence set is constructed by grid search over Θ, which becomes computationally prohibitive in high dimensions.
4. **Pre-specified estimands only.** The original framework assumes the estimand θ^* is chosen before seeing the data. If the researcher adaptively selects which subgroup or estimand to target based on the labeled data, the coverage guarantee breaks.

Taken together, the PPI framework solves the problem of valid inference with a fixed estimand and abundant unlabeled data, but it leaves unresolved the selective-inference problem created when the analyst uses labeled data to decide *which* estimand or subgroup to study. That gap is not peripheral: in many realistic uses of PPI, subgroup targeting is exactly the scientific goal. The remainder of this project therefore studies how to restore validity when subgroup choice is itself data-driven.

3 Extending PPI: Restoring Validity for Adaptive Subgroup Discovery

3.1 Motivation

The original PPI framework targets a single, pre-specified estimand $\theta^* = \mathbb{E}[Y]$ or a regression coefficient defined over the entire population. In practice, researchers often want to discover *where* a phenomenon is occurring: which subgroups defined by covariates X exhibit the largest effects, or where the model f is most or least reliable. This requires constructing confidence intervals for many subgroups simultaneously—and, critically, selecting which subgroups to examine based on the data itself.

Why adaptive selection breaks PPI. Suppose a researcher partitions the feature space $\mathcal{X}$ into K regions $R_1, \ldots, R_K$ and, for each region k, forms the PPI estimator $\hat{\theta}_k^{\mathrm{PP}}$ targeting $\theta_k^* = \mathbb{E}[Y \mid X \in R_k]$. Standard PPI intervals rely on asymptotic normality:

$$\sqrt{n}\big(\hat{\theta}_k^{\mathrm{PP}} - \theta_k^*\big) \xrightarrow{d} \mathcal{N}\big(0,\ \sigma_k^2\big), \qquad \sigma_k^2 = \mathrm{Var}\big(Y - f(X) \mid X \in R_k\big). \tag{2}$$

This is valid when the regions are *pre-specified*. However, when the regions are chosen *adaptively* based on the labeled data—for instance, by fitting a tree to the residuals $Y_i - f(X_i)$—the conditional expectation

$$\mathbb{E}\big[Y_i - f(X_i) \;\big|\; X_i \in R_k,\ \text{selection event}\big]$$

is no longer zero, because the selection event is a function of those same residuals. The rectifier is biased, and the coverage guarantee is destroyed. This is the classical "winner's curse" in selective inference.

Two strategies can restore validity: **sample splitting** (partition the labeled data into selection and inference sets) and **randomized inference** (add noise to Y for selection, then condition on the noise for inference). We develop both in the PPI setting and compare them. Following a suggestion by Prof. Barber, the randomized approach may be more powerful because it uses labeled data at all covariate locations rather than only a subset.

Scope of this section. Sections 3.1-3.5 develop the theoretical framework for both approaches. The sample-splitting branch is fully operationalized and empirically validated in Sections 4-5. The randomized-inference branch is a theoretical proposal with analytical predictions (e.g., the variance crossover in Section 3.4); its empirical evaluation is deferred to future work (Section 7.2).

3.2 Sample Splitting Baseline

3.2.1 Set-up & Notation

Let the labeled, "gold-standard" data be

$$\mathcal{D}_{\mathrm{lab}} = \{(X_i, Y_i)\}_{i=1}^n,$$

and let the unlabeled predictors be

$$\tilde{\mathcal{D}} = \{\tilde{X}_i\}_{j=1}^N,$$

where $N \gg n$. In addition, also suppose that f has been pre-trained, similar to the assumptions being made in the PPI framework. Now, we split $\mathcal{D}_{\mathrm{lab}}$ into two, disjoint datasets: $\mathcal{D}_{\mathrm{sel}}$ and $\mathcal{D}_{\mathrm{inf}}$, where $\mathcal{D}_{\mathrm{sel}}$ is only used for adaptive subgroup discovery, and $\mathcal{D}_{\mathrm{inf}}$ is used for inference.

Using $\mathcal{D}_{\mathrm{sel}}$, we fit a CART tree to the residuals $\varepsilon_i = Y_i - f(X_i)$, producing a partition $\{R_1, \ldots, R_K\}$ of the feature space.

Let $\rho \in (0, 1)$ be the proportion of values in $\mathcal{D}_{\mathrm{lab}}$. Then, when we assign both the inference-set and the unlabeled observations $\tilde{\mathcal{D}}$ to the same regions. For each region (in this case R_1, R_2), we compute a PPI

estimate. Following similar steps to the PPI framework, we compute

$$\begin{aligned}
\hat{\theta}_{R_i} &= \tilde{\theta}^f_{R_i} - \widehat{\mathbf{\Delta}}_{R_i} \\
\tilde{\theta}^f_{R_i} &= \frac{1}{N_{R_i}} \sum_{j:\tilde{X}_j \in R_i} f(\tilde{X}_j) \\
\widehat{\mathbf{\Delta}}_{R_i} &= \frac{1}{n_{\text{inf},R_i}} \sum_{j \in \mathcal{D}_{\text{inf}}: X_j \in R_i} (f(X_i) - Y_i)
\end{aligned}$$

The formulas are identical to standard PPI, applied region by region. The key gain is not a new estimator but the restored independence between adaptive region selection and rectifier estimation: because $\mathcal{D}_{\text{sel}} \perp \mathcal{D}_{\text{inf}}$, the rectifier in each discovered region is unbiased by construction.

3.2.2 Variance & Confidence Intervals

We can slightly tweak the results from section 2.4 to find the asymptotic variance of this estimator and a proposed confidence interval:

$$\widehat{\text{Var}}\left(\hat{\theta}_{R_i}\right) = \frac{\hat{\sigma}^2_{f-Y,R_i}}{n_{\text{inf},R_i}} + \frac{\hat{\sigma}^2_{f,R_i}}{N_{R_i}}$$

$$\text{Confidence Interval: } \boxed{\hat{\theta}_{R_i} \pm z_{1-\alpha/2}\widehat{\text{SE}}(\hat{\theta}_{R_i})}$$

where

$$\begin{aligned}
\hat{\sigma}^2_{f-Y,R_i} &= \widehat{\text{Var}}(Y - f(X) | X \in R_i) \\
\hat{\sigma}^2_{f,R_i} &= \widehat{\text{Var}}(f(\tilde{X}) | \tilde{X} \in R_i)
\end{aligned}$$

Now, let $\pi_i := \mathbb{P}[X \in R_i]$. Then, we can rewrite the denominators in terms of ρ and π:

$$\widehat{\text{Var}}\left(\hat{\theta}_{R_i}\right) = \frac{\hat{\sigma}^2_{f-Y,R_i}}{(1-\rho)n\pi_i} + \frac{\hat{\sigma}^2_{f,R_i}}{N\pi_i}$$

This way, we can clearly notice the effect of varying ρ: for larger ρ, we have more labeled observations in the "discovery" dataset, which will improve the quality of region selection. However, it comes at the cost of a wider confidence interval, since clearly the variance scales with the first term when we operate under the assumption of $N \gg n$. However, for lower ρ, we may not have good region selection. This ρ represents a form of bias-variance tradeoff.

3.2.3 Limitations of Sample Splitting

As with all other sample splitting methods, the main limitation is the loss of data. Given that one motivation for implementing PPI is a lack of "gold-standard" data, further splitting the data only serves to exacerbate this problem. In addition, there is no clear metric to determine what the best ρ value is, given that there is a tradeoff between confidence interval length and precision of region selection. The "best" choice of ρ is chosen by treating ρ as a tuning parameter and evaluating a grid of candidate values and comparing different factors such as interval width and selection accuracy.

3.3 Randomized Inference for PPI

An alternative strategy, inspired by the randomized-response literature and recent work on selective inference, avoids splitting entirely. The idea is to reveal a *noisy* version of Y for the selection step, then perform inference using the portion of Y that is conditionally independent of the noise.

Step 1: Noise injection. Generate independent noise $\varepsilon_i \sim \mathcal{N}(0, \tau^2)$ for each labeled point, and form

$$Z_i = Y_i + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Define the **selection information fraction**

$$\eta = \frac{\sigma_Y^2}{\sigma_Y^2 + \tau^2} \;\in\; (0, 1), \tag{3}$$

which controls how much of Y's signal is revealed through Z.

Step 2: Adaptive selection. Use $\{(X_i, Z_i)\}_{i=1}^n$ together with the unlabeled predictions $\{f(\tilde{X}_j)\}_{j=1}^N$ to adaptively partition $\mathcal{X}$ into regions $R_1, \ldots, R_K$. Any selection procedure is permitted—decision trees on the noisy residuals $Z_i - f(X_i)$, clustering, thresholding the imputed means, etc.—because validity depends only on the independence structure in Step 3, not on the selection algorithm.

Step 3: Independent inference residual. Define the **inference residual**

$$W_i = Y_i - \eta\, Z_i. \tag{4}$$

A direct calculation gives $\mathrm{Cov}(W_i, Z_i) = \sigma_Y^2 - \eta(\sigma_Y^2 + \tau^2) = 0$ by the definition of η. Under Gaussian Y and ε, zero covariance implies independence: $W_i \perp Z_i$. Therefore W_i is independent of the entire selection event (which is a function of $Z_1, \ldots, Z_n$ and the unlabeled data).

The variance of W_i is

$$\mathrm{Var}(W_i) = \mathrm{Var}(Y_i) - 2\eta\,\mathrm{Cov}(Y_i, Z_i) + \eta^2\,\mathrm{Var}(Z_i) = (1 - \eta)\,\sigma_Y^2. \tag{5}$$

Step 4: PPI estimator conditional on Z. For a selected region R, write $n_R = |\{i : X_i \in R\}|$ (all labeled points in R) and $N_R = |\{j : \tilde{X}_j \in R\}|$. Decompose the PPI rectifier:

$$f(X_i) - Y_i = \big[f(X_i) - \eta\, Z_i\big] - W_i. \tag{6}$$

Conditional on Z, the term $f(X_i) - \eta Z_i$ is a known constant. The randomness in the rectifier comes entirely from W_i, which is independent of the selection.

The **randomized PPI estimator** for $\theta_R^* = \mathbb{E}[Y \mid X \in R]$ is:

$$\hat{\theta}_R^{\mathrm{rand}} \;=\; \underbrace{\frac{1}{N_R} \sum_{j:\, \tilde{X}_j \in R} f(\tilde{X}_j)}_{\tilde{\theta}_R^f} \;+\; \frac{1}{n_R} \sum_{i:\, X_i \in R} \Big[W_i - \big(f(X_i) - \eta\, Z_i\big)\Big]. \tag{7}$$

Conditional on Z, this is a sum of independent terms (the W_i's and the unlabeled data), so the CLT applies:

$$\mathrm{Var}\big(\hat{\theta}_R^{\mathrm{rand}} \mid Z\big) \;\approx\; \frac{(1 - \eta)\,\sigma_Y^2}{n_R} \;+\; \frac{\sigma_f^2}{N_R}. \tag{8}$$

A standard confidence interval can then be formed using this conditional variance estimate, and its coverage is valid because the inference residuals are independent of the selection event.

In particular, we note that this estimator is unbiased. Taking expectations over W conditional on Z and noting $\mathbb{E}[W_i] = \mathbb{E}[Y_i] - \eta\,\mathbb{E}[Z_i] = (1 - \eta)\,\theta_R^*$:

$$\mathbb{E}\big[\hat{\theta}_R^{\mathrm{rand}} \mid Z\big] = \tilde{\theta}_R^f + (1 - \eta)\,\theta_R^* - \big[\bar{f}_R - \eta\,\bar{Z}_R\big] = \theta_R^* + (\tilde{\theta}_R^f - \bar{f}_R) + \eta(\bar{Z}_R - \theta_R^* - \bar{\varepsilon}_R),$$

where $\bar{f}_R = n_R^{-1} \sum f(X_i)$ and $\bar{Z}_R = n_R^{-1} \sum Z_i$. The second and third terms vanish in expectation over Z and the unlabeled data: $\mathbb{E}[\bar{Z}_R] = \theta_R^*$ and $\mathbb{E}[\tilde{\theta}_R^f] = \mathbb{E}[f(X)]$, $\mathbb{E}[\bar{f}_R] = \mathbb{E}[f(X)]$. So the estimator is unconditionally unbiased for θ_R^*.

3.4 Comparing the Two Approaches

To place sample splitting and randomized inference on comparable footing, we adopt the heuristic calibration $\eta = \rho$, which aligns the fraction of signal allocated to selection in both approaches. This is a first-order comparison device rather than a canonical equivalence, since the two methods use information in structurally different ways, but it provides a useful baseline for variance comparison.

For a region R with $P(X \in R) = \pi$ and $n_R \approx n\pi$, the rectifier-variance contributions are:

$$V_{\text{split}}(\rho) = \frac{\sigma^2_{f\text{-}Y}}{(1-\rho)\, n\pi}, \tag{9}$$

$$V_{\text{rand}}(\eta = \rho) = \frac{(1-\rho)\, \sigma^2_Y}{n\pi}. \tag{10}$$

(The $\sigma^2_f/(N\pi)$ term is identical and cancels.)

Specifically, we study when does randomized inference win. Setting $V_{\text{rand}} < V_{\text{split}}$ and simplifying:

$$(1-\rho)^2 \;<\; \frac{\sigma^2_{f\text{-}Y}}{\sigma^2_Y}. \tag{11}$$

If the predictor f explains an R^2 fraction of the outcome variance (so $\sigma^2_{f\text{-}Y} \approx (1-R^2)\sigma^2_Y$), this becomes $(1-\rho)^2 < 1 - R^2$, i.e.,

$$\rho \;>\; 1 - \sqrt{1-R^2}. \tag{12}$$

This yields a non-obvious and practically useful conclusion:

- **Poor model** (R^2 low): $\sigma^2_{f\text{-}Y} \approx \sigma^2_Y$, so the threshold (12) is near zero. The randomized approach wins for almost any $\rho > 0$ because the per-point variance penalty is mild and the spatial-coverage advantage dominates. Concretely, for $R^2 = 0.1$, the threshold is $\rho > 0.051$.
- **Good model** (R^2 high): $\sigma^2_{f\text{-}Y} \ll \sigma^2_Y$, so sample splitting leverages the model quality through the smaller rectifier variance $\sigma^2_{f\text{-}Y}$. The randomized approach's per-point variance $(1-\eta)\sigma^2_Y$ does not benefit from model accuracy. For $R^2 = 0.9$, the threshold is $\rho > 0.684$—randomized inference only wins if the majority of the data is allocated to selection.
- **Rare subgroups** (π small): The randomized approach uses $n_R \approx n\pi$ labeled points in R; splitting uses only $(1-\rho)n\pi$. When π is small, even a modest ρ can push $n_{\text{inf},R}$ below the threshold needed for a reliable CLT approximation, making the randomized approach preferable for its better spatial coverage.

3.5 Optimal Noise Calibration

The choice of η (equivalently τ^2) in the randomized scheme mirrors the choice of ρ in sample splitting. Both control a selection-inference tradeoff:

As η increases from 0 toward 1, the selection view Z_i becomes more informative (because less noise is added), enabling the selection procedure to identify stronger signals. Simultaneously, the independent residual $W_i = Y_i - \eta Z_i$ retains less variance: $\text{Var}(W_i) = (1-\eta)\sigma^2_Y$, which shrinks toward zero. In the extreme, $\eta \to 1$ gives perfect selection information but a degenerate inference residual, while $\eta \to 0$ preserves the full outcome for inference but provides no guidance for selection. The optimal η^* balances these two forces.

Define the **discovery power** for a region R with signal $\mu_R = \theta^*_R$ as the probability that the PPI confidence interval excludes zero. The non-centrality parameter of the randomized PPI test is

$$\lambda_R(\eta) = \frac{|\mu_R|}{\sqrt{(1-\eta)\, \sigma^2_Y/n_R + \sigma^2_f/N_R}}. \tag{13}$$

Meanwhile, the *quality* of the selection step depends on η: larger η reveals more signal in Z, enabling the selection procedure to identify regions with larger $|\mu_R|$ or smaller σ_R^2. Let $\mu_R(\eta)$ denote the expected signal strength in the selected region as a function of η (which increases in η).

The optimal η^* maximizes $\lambda_R(\eta)$ accounting for both effects. In general this requires specifying the signal landscape, but a key structural insight holds: as η increases from 0, $\mu_R(\eta)$ increases (better selection) while $(1-\eta)$ decreases (tighter inference)—the numerator and denominator of λ_R both improve. But $\mu_R(\eta)$ eventually saturates (once selection is good enough to find the best region), while the inference variance $(1-\eta)\sigma_Y^2$ continues to shrink toward zero, collapsing the confidence interval width but also the signal available in W. The optimal η^* balances these forces.

We will characterize η^* empirically via simulation across varying signal strengths, model qualities, and subgroup structures.

3.6 Simulation Plan

We will evaluate the randomized PPI approach against sample splitting across a grid of scenarios:

1. **Data Generation**: Features $X_i \in \mathbb{R}^d$ with $d = 3$, with labeled points generated via $Y_i = f(X_i) + \epsilon_i$. Furthermore, we will generate Y in such a way that there are clear, distinguishable regions. **Selection Procedure:** For the sample-splitting experiments implemented in Sections 4-5, we fit a CART tree to the residuals $Y_i - f(X_i)$ computed on $\mathcal{D}_{\text{sel}}$. For the randomized approach (not implemented empirically in this report), selection would use the noisy residuals $Z_i - f(X_i)$, where $Z_i = Y_i + \varepsilon_i$ as defined in Section 3.3.
2. **Output**: For each selected region, we compute a PPI confidence interval, interval width, and selection accuracy
3. **Sweep over η and ρ**: For each scenario, we vary the tuning parameter to trace the coverage-width tradeoff, and empirically identify the optimal η and ρ, η^* and ρ^* respectively.

3.7 Questions for the Simulation Study

The simulation study implemented in Sections 4–5 addresses the following questions for the **sample-splitting branch**:

1. **Coverage validity:** Does sample splitting maintain near-nominal coverage across the ρ grid, confirming that the selection-inference separation successfully decouples adaptive region choice from PPI estimation?

2. **Width-selection tradeoff:** How does the confidence-interval width grow with ρ, and does the selection procedure benefit meaningfully from a larger $\mathcal{D}_{\text{sel}}$?

3. **Optimal allocation:** What is the empirically optimal ρ^* minimizing width subject to valid coverage?

4. **Multi-region FDR control:** Does applying BH to region-wise PPI p-values control the false discovery rate under sample splitting?

The variance crossover condition (equation 12) and the spatial-coverage advantage of the randomized approach are *analytical predictions* from Section 3.4; empirical validation of the randomized branch is deferred to future work.

4 Simulation Study: Single-Region Validity and Efficiency

4.1 Setup

4.1.1 Data Generating Process

In each Monte Carlo replication, we generated three independent datasets: an external pretraining sample of size $n_{\text{pre}} = 2000$, a labeled sample of size $n = 300$, and an unlabeled sample of size $N = 5000$. Covariates were drawn as $X_1, X_2 \overset{\text{iid}}{\sim} N(0,1)$ and $X_3 \sim \text{Bernoulli}(0.5)$. For simplicity of this simulation, outcomes were generated from

$$Y = 1 + 1.5X_1 - X_2 + 0.8X_3 + 1.2\,\mathbf{1}\{X_1 > 0, X_2 > 0\} + \varepsilon, \quad \varepsilon \sim N(0,1).$$

The interaction term induces a genuine signal-bearing subgroup $R^* = \{X_1 > 0, X_2 > 0\}$, whose population mass is approximately 0.25. In this region, the conditional mean of Y is elevated relative to the rest of the feature space, providing a known ground truth subgroup against which we can evaluate whether the adaptive selection procedure successfully recovers the signal.

The prediction function f was fit by linear regression on an independent pretraining sample, ensuring that the predictor is external to both the labeled and unlabeled datasets as required by the PPI framework. The labeled sample was used for subgroup selection and inference, while the unlabeled sample was drawn from the same covariate distribution but contributed only through the imputed predictions $f(\tilde{X})$ in the PPI estimator.

Because the true outcome model includes a nonlinear interaction term $\mathbf{1}\{X_1 > 0, X_2 > 0\}$ while the prediction function f is fit using a simple linear regression model, we have intentionally misspecified the prediction model. This misspecification limits the ability of f to capture the interaction structure in the data and therefore reduces the efficiency gains that PPI can achieve relative to a well-specified prediction model.

This design choice also highlights an important limitation of the PPI framework: **the magnitude of the efficiency gains and power of the PPI framework is heavily dependent on model quality**. When the prediction model is misspecified, the variance of the rectifier term increases, reducing the improvement in interval precision that PPI can provide.

4.1.2 Region Selection Procedure

To identify candidate subgroups, we use a simple regression tree fit to the residuals of the prediction model. Let

$$\varepsilon_i = Y_i - f(X_i)$$

denote the residual for observation i in the labeled dataset.

Among the K leaves produced by the tree, we define the selected region R_b as the leaf with the largest absolute mean residual on the selection set:

$$R_b = \arg\max_{R_k} \left| \frac{1}{n_{\text{sel},R_k}} \sum_{i \in \mathcal{D}_{\text{sel}}:\, X_i \in R_k} (Y_i - f(X_i)) \right|.$$

Because this criterion uses only $\mathcal{D}_{\text{sel}}$, the choice of which leaf to report is independent of the inference data $\mathcal{D}_{\text{inf}}$, preserving the selection-inference separation that sample splitting requires. The coverage, width, and Jaccard metrics in Sections 4.2-4.4 are then computed for R_b using $\mathcal{D}_{\text{inf}}$ only.

To allow the selection procedure to capture interactions between covariates while still maintaining reasonably sized regions for inference, we fit a shallow CART tree with maximum depth two. This produces a partition of the feature space into at most four regions defined by sequential threshold splits along the covariates. Allowing two levels of splitting enables the selection procedure to approximate the true subgroup structure induced by the interaction between X_1 and X_2, while avoiding excessively small leaves that could lead to unstable variance estimates in the inference stage.

Once the tree is fitted on D_{sel}, the resulting partition is applied to both the inference dataset D_{inf} and the unlabeled dataset $\tilde{D}$. Each observation is assigned to the corresponding region R_k according to the learned split rule.

For each region R_k, we then compute the PPI estimator

$$\hat{\theta}_{R_k} = \tilde{\theta}^f_{R_k} + \frac{1}{n_{\text{inf},R_k}} \sum_{i \in D_{\text{inf}}: X_i \in R_k} (Y_i - f(X_i)).$$

4.1.3 PPI Estimation Within Regions

After the region-selection step produces a partition of the feature space into regions $R_1, \ldots, R_K$, we estimate the conditional mean

$$\theta^*_R = \mathbb{E}[Y \mid X \in R]$$

for each selected region R using the prediction-powered inference (PPI) estimator.

For a given region R, let

$$n_{\text{inf},R} = |\{i \in D_{\text{inf}} : X_i \in R\}|$$

denote the number of labeled inference observations in that region, and

$$N_R = \left|\{j : \tilde{X}_j \in R\}\right|$$

the number of unlabeled observations assigned to the same region.

The PPI estimator combines the average prediction over the unlabeled sample with a correction term based on labeled residuals:

$$\hat{\theta}_R = \tilde{\theta}^f_R + \frac{1}{n_{\text{inf},R}} \sum_{i \in D_{\text{inf}}: X_i \in R} \big(Y_i - f(X_i)\big),$$

where

$$\tilde{\theta}^f_R = \frac{1}{N_R} \sum_{j: \tilde{X}_j \in R} f(\tilde{X}_j).$$

The variance of the estimator is estimated using the plug-in formula

$$\widehat{\text{Var}}(\hat{\theta}_R) = \frac{\widehat{\sigma}^2_{Y-f,R}}{n_{\text{inf},R}} + \frac{\widehat{\sigma}^2_{f,R}}{N_R},$$

where $\widehat{\sigma}^2_{Y-f,R}$ is the sample variance of $Y_i - f(X_i)$ among inference observations in region R, and $\widehat{\sigma}^2_{f,R}$ is the sample variance of $f(\tilde{X}_j)$ among unlabeled observations in that region.

A $100(1-\alpha)\%$ confidence interval for θ^*_R is then constructed as

$$\hat{\theta}_R \pm z_{1-\alpha/2} \sqrt{\widehat{\text{Var}}(\hat{\theta}_R)}.$$

4.2 Coverage Validation

We assess the statistical validity of the proposed procedure by examining the empirical coverage of the resulting confidence intervals through a Monte Carlo experiment with $B = 1000$ independent replications. We use a nominal level of $\alpha = 0.10$ (90% coverage) in the single-region experiment to make finite-sample deviations from nominal coverage more visible at $n = 300$. Section 5 reverts to $\alpha = 0.05$ (95% coverage) for the multi-region pipeline, which is the standard used in practice. In each replication, the pretraining, labeled, and unlabeled datasets are generated according to the data-generating process described in the previous subsection.

For each replication, the labeled dataset is randomly partitioned into a selection sample D_{sel} and an inference sample D_{inf}. The region-selection algorithm described above is applied to D_{sel}, producing a partition of the feature space into regions $R_1, \ldots, R_K$. The same region definitions are then applied to the inference dataset and the unlabeled dataset. For each region R_k, we compute the PPI estimator and its corresponding confidence interval using the inference observations and the unlabeled predictions.

Because the regions are selected adaptively from the data, the inferential target

$$\theta^*_R = \mathbb{E}[Y \mid X \in R]$$

depends on the realized region R in each replication. To obtain a reference value for this quantity, we generate a large independent evaluation dataset of size $n_{\text{eval}} = 200{,}000$ from the same data-generating process and estimate

$$\hat{\theta}_{R,\text{eval}} = \frac{\sum_{m=1}^{n_{\text{eval}}} Y_m^{\text{eval}} \mathbf{1}\{X_m^{\text{eval}} \in R\}}{\sum_{m=1}^{n_{\text{eval}}} \mathbf{1}\{X_m^{\text{eval}} \in R\}}.$$

Coverage is then defined as the proportion of Monte Carlo replications in which the constructed confidence interval contains the approximated true parameter. Formally,

$$\widehat{\text{Coverage}} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\{\hat{\theta}_{R_b,\text{eval}} \in CI_b\},$$

where CI_b denotes the confidence interval computed in replication b for the corresponding selected region R_b. Under valid inference, the empirical coverage should be close to the nominal confidence level.

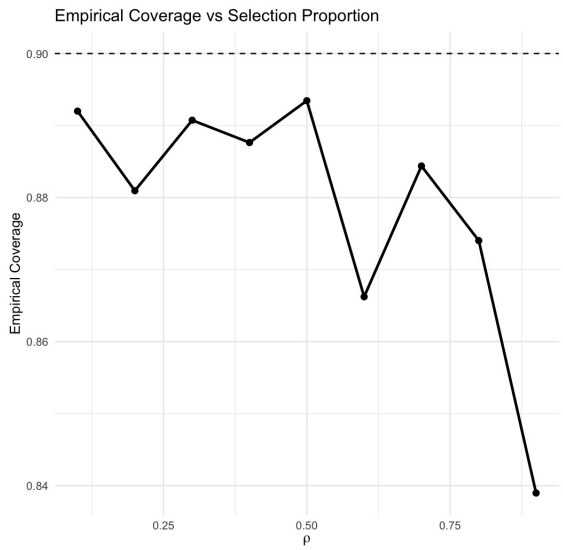


Table 1 and Figure 4.2 report the empirical coverage of the PPI confidence intervals across values of the selection proportion ρ. The nominal coverage level is 0.90, which is indicated by the dashed horizontal line in Figure 4.2. For small and moderate values of ρ, empirical coverage remains close to the nominal level, ranging between approximately 0.88 and 0.89. However, as ρ increases further, coverage begins to decline, reaching approximately 0.84 when $\rho = 0.9$.

This pattern reflects the fundamental tradeoff induced by sample splitting. Increasing ρ allocates more labeled observations to the selection stage, leaving fewer observations available for inference. Since the PPI confidence interval relies on the inference sample to estimate the rectifier term and its variance, a smaller inference sample leads to less stable variance estimates and deterioration of coverage in finite samples.

In this design, the selection benefit of increasing ρ is negligible: Jaccard similarity stays between 0.19 and 0.23 across the grid (Table 1). Meanwhile, coverage drops from 0.89 to 0.84 as ρ goes from 0.1 to 0.9, driven entirely by the shrinking inference sample. The shallow CART tree, not the selection sample size, is the binding constraint on subgroup recovery here.

4.3 Interval Width Comparison

In addition to coverage, we evaluate the efficiency of the proposed procedure by examining the width of the resulting confidence intervals. For each replication and selected region R, we record the interval length

$$\text{Width}_R = \text{Upper}_R - \text{Lower}_R.$$

We report the average interval width across Monte Carlo replications,

$$\widehat{\text{Width}} = \frac{1}{B}\sum_{b=1}^{B} \text{Width}_{R_b},$$

where R_b denotes the region selected in replication b. Smaller values correspond to more precise inference.

We compare interval widths across different values of the selection proportion ρ, which determines the fraction of labeled data used for subgroup discovery in the sample-splitting procedure. Increasing ρ improves the quality of region selection but reduces the size of the inference sample, potentially leading to wider confidence intervals. The resulting curves illustrate the tradeoff between selection accuracy and estimation precision.

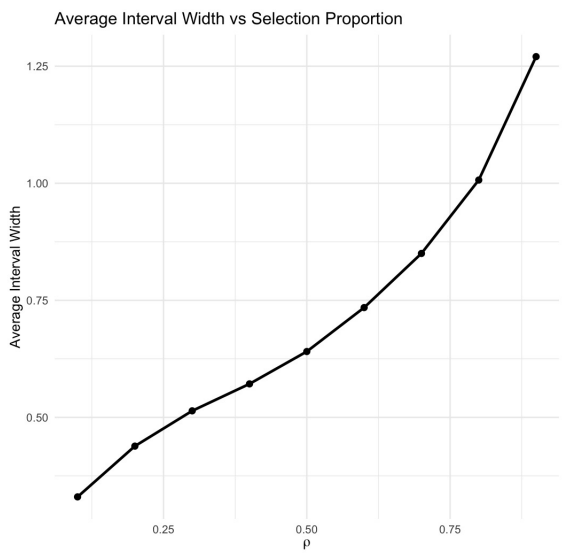


Figure 4.3 and Table 1 show the average width of the PPI confidence intervals across values of ρ. The interval width increases monotonically as ρ increases, rising from approximately 0.33 when $\rho = 0.1$ to over 1.27 when $\rho = 0.9$.

This pattern follows directly from the variance expression

$$\widehat{\text{Var}}(\hat{\theta}_R) = \frac{\sigma^2_{f-Y,R}}{(1-\rho)n\pi_R} + \frac{\sigma^2_{f,R}}{N\pi_R},$$

where $\pi_R = \mathbb{P}(X \in R)$ denotes the population mass of the region. In our setting $N \gg n$, so the second term is negligible and the dominant contribution to the variance arises from the rectifier term.

As ρ increases, fewer labeled observations remain in the inference sample D_{inf}, whose expected size within region R is approximately $(1-\rho)n\pi_R$. Consequently, the variance of the rectifier term grows approximately at rate $(1-\rho)^{-1}$. This inflation in variance directly translates into wider confidence intervals, illustrating the loss of inference precision when too large a fraction of the labeled data is allocated to subgroup discovery.

4.4 Selection Accuracy

Besides evaluating coverage and interval width, we assess how well the region-selection procedure identifies the true signal-bearing subgroup. In the data-generating process, the ground-truth subgroup is

$$R^* = \{X_1 > 0, X_2 > 0\}.$$

For each Monte Carlo replication, the selection procedure produces a region R_b. To quantify the similarity between the discovered region and the true subgroup, we compute the Jaccard similarity coefficient,

$$\text{Jaccard}(R_b, R^*) = \frac{|R_b \cap R^*|}{|R_b \cup R^*|}.$$

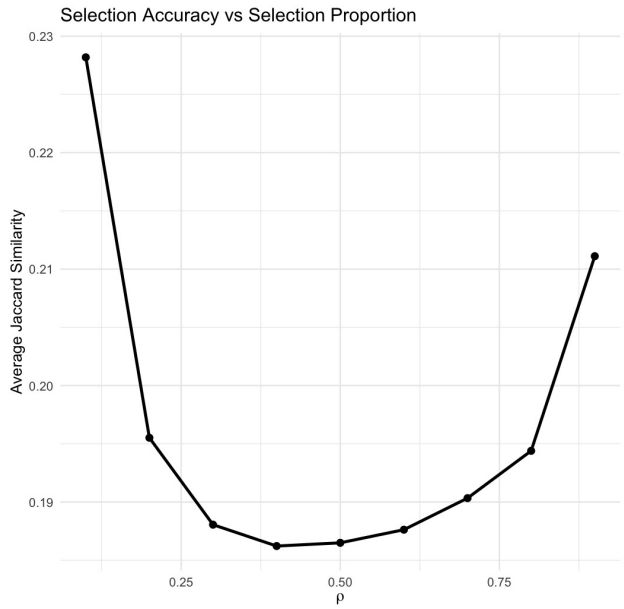
Because both regions are subsets of the feature space, we approximate these quantities using the large evaluation dataset described in the previous subsection. Specifically, we compute

$$\widehat{\text{Jaccard}}(R_b, R^*) = \frac{\sum_{m=1}^{n_{\text{eval}}} \mathbf{1}\{X_m^{\text{eval}} \in R_b \cap R^*\}}{\sum_{m=1}^{n_{\text{eval}}} \mathbf{1}\{X_m^{\text{eval}} \in R_b \cup R^*\}}.$$

We then report the average Jaccard similarity across Monte Carlo replications,

$$\widehat{\text{Accuracy}} = \frac{1}{B} \sum_{b=1}^{B} \widehat{\text{Jaccard}}(R_b, R^*).$$

Higher values indicate that the discovered regions more closely match the true signal-bearing subgroup.



The results are shown in Figure 4.4 and Table 1. Jaccard similarity ranges from 0.18 to 0.23–low in absolute terms, meaning the shallow CART tree does not recover the true interaction subgroup $R^* = \{X_1 > 0, X_2 > 0\}$ well. This is not unexpected: a depth-2 axis-aligned tree has limited capacity to approximate a two-dimensional rectangular region, and is further constrained by small selection samples at high ρ.

Importantly, the flatness of Jaccard across ρ confirms that the tree depth, not the selection sample size, is the binding constraint on recovery. As a consequence, the main effect of increasing ρ is not improved subgroup discovery but rather the loss of inference precision. This reinforces the paper's primary focus: testing whether sample splitting preserves coverage validity, not whether the selector finds the best subgroup.

4.5 Optimal ρ^*

Finally, we investigate the optimal allocation of information between the subgroup discovery and inference steps. In the sample-splitting framework, this allocation is controlled by the selection proportion ρ. (In the randomized inference framework introduced later, the analogous role is played by the information fraction η.)

For the sample-splitting experiment, we sweep across a grid of values

$$\rho \in \{0.1, 0.2, \ldots, 0.9\}.$$

For each value of ρ, we estimate empirical coverage and average interval width using the Monte Carlo procedure described above.

Because empirical coverage is estimated with Monte Carlo error, we define a split ratio as *admissible* if its empirical coverage is within Monte Carlo uncertainty of the nominal level. With $B = 1000$ replications, the standard error of the coverage estimate is approximately $\sqrt{0.90 \cdot 0.10/1000} \approx 0.01$, so we treat any ρ with $\widehat{\text{Coverage}}(\rho) \geq 0.88$ as admissible. Among admissible values, we select the one with minimum average width:

$$\rho^* = \arg\min_{\rho} \widehat{\text{Width}}(\rho) \quad \text{subject to} \quad \widehat{\text{Coverage}}(\rho) \geq 0.90 - 2\,\text{SE}_{\text{MC}}.$$

Under this rule, $\rho = 0.1$ is preferred: it gives the smallest width (0.33) while remaining statistically indistinguishable from nominal coverage at the current number of replications.

The empirical results reported in Table 1 show that the smallest interval widths occur for small values of ρ, particularly between $\rho = 0.1$ and $\rho = 0.3$, where empirical coverage remains close to the nominal 0.90 level while interval widths are substantially smaller than for larger values of ρ. As ρ increases, the inference sample $(1 - \rho)n$ becomes smaller, leading to wider confidence intervals and a gradual deterioration of coverage.

The empirical picture is simpler than the generic tradeoff language might suggest: in this DGP, the "selection benefit" of larger ρ is weak (Jaccard is flat), while the "inference cost" is strong (width nearly quadruples from $\rho = 0.1$ to $\rho = 0.9$). As a result, the smallest feasible ρ dominates. Whether this pattern persists under stronger interaction signals or deeper trees–where selection quality might respond more to ρ–is an interesting question for future work.

ρ	Mean Coverage	Mean Interval Width	Mean Jaccard Similarity
0.1	0.892	0.330	0.228
0.2	0.881	0.439	0.196
0.3	0.891	0.514	0.188
0.4	0.888	0.572	0.186
0.5	0.893	0.641	0.186
0.6	0.866	0.734	0.188
0.7	0.884	0.850	0.190
0.8	0.874	1.010	0.194
0.9	0.839	1.270	0.211

Table 1: Monte Carlo simulation results summarizing empirical coverage, average interval width, and selection accuracy for different values of the sample-splitting proportion ρ.

Summary. The single-region sample-splitting experiment establishes three findings. First, the procedure maintains near-nominal coverage at small ρ, confirming that the selection–inference separation works as intended. Second, interval width is driven almost entirely by the inference sample size: it scales as $(1 - \rho)^{-1}$ and grows from 0.33 to 1.27 across the ρ grid. Third, selection accuracy is essentially flat at Jaccard $\approx$ 0.19–0.23, indicating that the shallow CART tree rather than the selection sample size is the bottleneck for subgroup recovery. Together, these results make a strong case for low ρ in this regime. They also motivate the extension in Section 5: since the tree returns multiple leaves, a natural next step is to test and report all of them with appropriate multiplicity control rather than selecting a single leaf.

5 Extension: Multiple-Region Discovery and Selective Reporting under Sample Splitting (Valid Adaptive Splitting)

Sections 3 and 4 address validity for an adaptively chosen subgroup. In practice, however, a tree-based discovery procedure returns a partition with multiple candidate leaves, and the analyst may wish to test and report several of them. This creates a distinct inferential problem: even if each region-wise PPI interval is marginally valid under sample splitting, naive reporting across all discovered regions ignores multiplicity. In this section, we extend the sample-splitting framework from Section 3.2 to the multiple-region setting by combining region-wise PPI p-values with BH selection and selective interval adjustment.

5.1 Procedure

Stage 1: Selection Using D_{sel}, we fit a shallow CART tree to the residuals

$$\hat{\varepsilon}_i = Y_i - f(X_i), \qquad i \in D_{\text{sel}},$$

which partitions the feature space into regions

$$\mathcal{P} = \{R_1, \ldots, R_K\}.$$

As in Section 3, the key property is that $\mathcal{P}$ is a function of D_{sel} only, and is therefore independent of D_{inf}. As in Section 4, we use a depth-2 CART tree. The difference in this section is not the partition complexity but that we retain and test all discovered leaves rather than selecting only a single leaf for reporting.

Stage 2: PPI Inference We assign the inference sample D_{inf} and the unlabeled sample $\widetilde{D}$ to the learned regions $R_1, \ldots, R_K$. For each region R_k, we then compute the sample-splitting PPI estimator $\widehat{\theta}^{\text{PP}}_{R_k}$ and its standard error $\widehat{\text{SE}}_{R_k}$ exactly as in Section 3.2, now applied separately within each discovered region.

Thus, for each $k = 1, \ldots, K$, we test

$$H_{0,k} : \theta^*_{R_k} = 0$$

using the usual two-sided normal approximation

$$p^{\text{PP}}_k = 2\left[1 - \Phi\left(\left|\widehat{\theta}^{\text{PP}}_{R_k} \Big/ \widehat{\text{SE}}_{R_k}\right|\right)\right]. \tag{14}$$

Stage 3: Multiple-region selection and selective reporting The new issue in this section is that the tree may produce several candidate regions, so naive reporting across all leaves would ignore multiplicity. To address this, we apply the Benjamini–Hochberg (BH) procedure at level α_{BH} to the collection of region-wise PPI p-values

$$p^{\text{PP}}_1, \ldots, p^{\text{PP}}_K.$$

Let $\mathcal{S} \subseteq \{1, \ldots, K\}$ denote the set of BH-selected regions.

For each selected region $k \in \mathcal{S}$, we report an FCR-adjusted confidence interval based on the same region-wise estimator and standard error from Stage 2:

$$C^{\text{PP}}_k = \widehat{\theta}^{\text{PP}}_{R_k} \pm z_{\,1-\alpha|\mathcal{S}|/(2K)}\, \widehat{\text{SE}}_{R_k}. \tag{15}$$

Here α is the nominal per-interval level. The factor $|\mathcal{S}|/K$ widens the reported intervals to account for selective reporting across the discovered regions.

5.2 Validity

The validity of this pipeline follows from two observations: First, since the partition $\mathcal{P} = \{R_1, \ldots, R_K\}$ depends only on $\mathcal{D}_{\text{sel}}$, and $\mathcal{D}_{\text{sel}} \perp \mathcal{D}_{\text{inf}}$ by construction, the rectifier residuals $\{f(X_i) - Y_i : i \in \mathcal{D}_{\text{inf}},\, X_i \in R_k\}$ are conditionally i.i.d. given $\mathcal{P}$. In particular,

$$\mathbb{E}[f(X_i) - Y_i \mid X_i \in R_k,\, \mathcal{D}_{\text{sel}}] = \mathbb{E}[f(X_i) - Y_i \mid X_i \in R_k], \tag{16}$$

so the empirical rectifier $\hat{\Delta}_{R_k}$ remains unbiased in each adaptively-discovered region. This is the same conditional independence argument used in Section 3.2, now applied region-by-region.

Second, conditional on $\mathcal{P}$, the region-wise p-values are built from disjoint subsets of the inference and unlabeled data. This suggests a dependence structure under which BH should be appropriate; under independence or PRDS-type conditions, BH controls FDR at level $\alpha_{\mathrm{BH}} \cdot K_0/K$, where K_0 is the number of truly null regions. We do not prove the required dependence condition formally, and instead evaluate FDR control empirically in Section 5.3.

Together, these two properties ensure that (i) PPI confidence intervals in each discovered region have valid marginal coverage, and (ii) applying BH to the collection of PPI p-values controls FDR.

5.3 Results

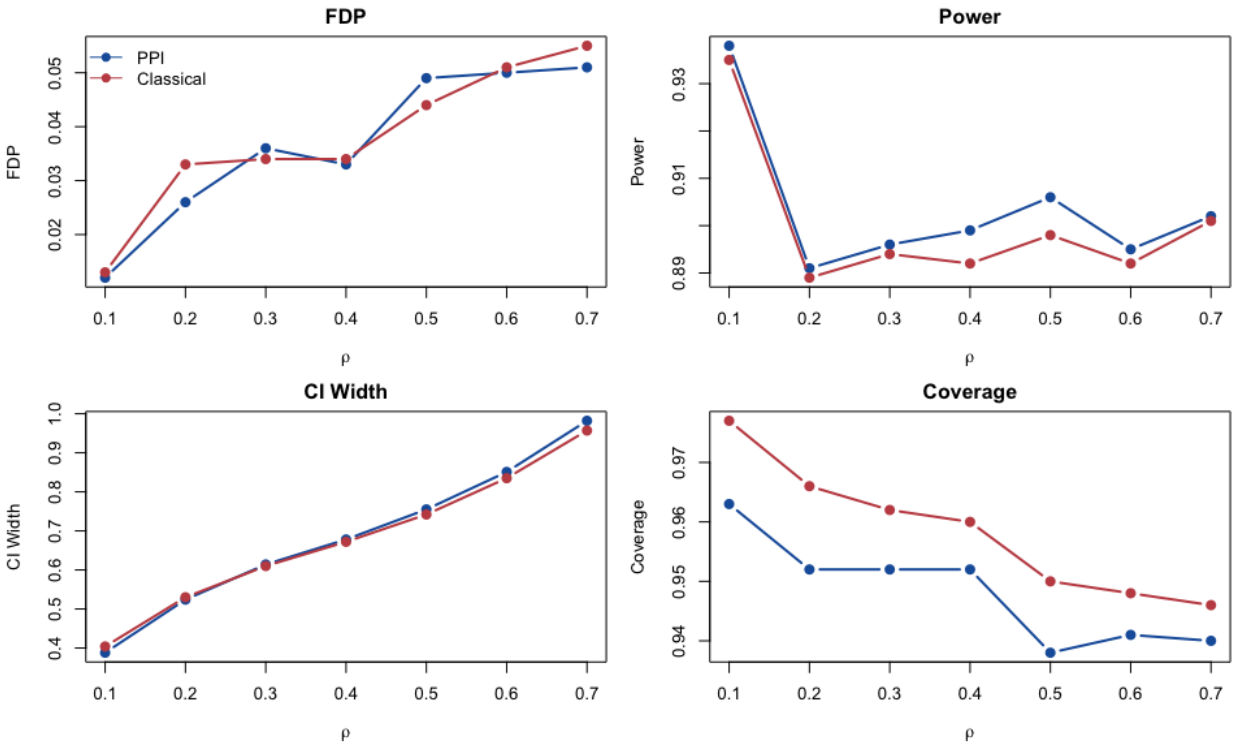


Figure 1: FDR, power, CI width, and coverage of the valid adaptive splitting pipeline

Table 2: Simulation results for the valid adaptive splitting + BH pipeline. FDP = false discovery proportion; Pwr = power (fraction of non-null regions discovered); Wid = average CI width; Cov = marginal coverage. BH level $\alpha_{\mathrm{BH}} = 0.10$; CI level $\alpha = 0.05$; $B = 500$ replications.

	PPI				Classical			
ρ	FDP	Pwr	Wid	Cov	FDP	Pwr	Wid	Cov
0.1	0.012	0.938	0.388	0.963	0.013	0.935	0.404	0.977
0.2	0.026	0.891	0.524	0.952	0.033	0.889	0.530	0.966
0.3	0.036	0.896	0.614	0.952	0.034	0.894	0.610	0.962
0.4	0.033	0.899	0.678	0.952	0.034	0.892	0.672	0.960
0.5	0.049	0.906	0.755	0.938	0.044	0.898	0.742	0.950
0.6	0.050	0.895	0.851	0.941	0.051	0.892	0.835	0.948
0.7	0.051	0.902	0.982	0.940	0.055	0.901	0.957	0.946

5.3.1 FDR Control

Across all split ratios, the mean false discovery proportion (mean FDP, which serves as an empirical estimate of FDR) remains well below the BH target of 0.10, ranging from 0.012 ($\rho = 0.1$) to 0.051 ($\rho = 0.7$). These results are consistent with BH controlling FDR in this sample-splitting PPI pipeline. Because the relevant dependence condition is not formally proved here, we interpret Table 2 as empirical support rather than a general theorem. The slight increase in FDR at larger ρ reflects the reduced inference sample size, which degrades the CLT approximation underlying the PPI p-values.

5.3.2 Coverage and Width

PPI coverage remains near or above the nominal 0.95 level for $\rho \leq 0.4$, with a mild decline to $\sim$0.94 at $\rho \geq 0.5$. Classical coverage is consistently slightly higher as the classical estimator does not involve the prediction model f, so its variance estimate is more stable in small subgroups.

The PPI and classical CI widths are nearly identical across all ρ. Although the one-region variance heuristic from Section 6.2 predicts a $\sqrt{0.78} \approx 0.88$ width ratio (i.e., $\sim$12% reduction) under $R^2 \approx 0.22$, this gain does not materialize cleanly in the multi-region selective pipeline: PPI is marginally narrower at low ρ (e.g., 0.388 vs. 0.404 at $\rho = 0.1$) but comparable or slightly wider at larger ρ values. The FCR adjustment, small per-region sample sizes, and leaf-wise variance estimation all erode the theoretical advantage. A stronger predictor would be needed to produce a visible PPI benefit in this pipeline.

5.3.3 Power and Selection-Inference Tradeoff

Power is highest at $\rho = 0.1$ (PPI: 0.938, Classical: 0.935) and declines at intermediate ρ before stabilizing around 0.90 for $\rho \geq 0.3$. This pattern reflects the tradeoff from Section 3.2.3: at low ρ, the inference sample is large enough to detect most non-null regions even with a crude partition, while at high ρ the larger selection sample may improve partition quality, though Section 5 does not directly measure this effect (e.g., via Jaccard similarity). The stabilization of power at $\rho \geq 0.3$ is consistent with a partial compensation, but a definitive claim would require an explicit recovery metric

For this DGP, the optimal split ratio is $\rho^* \approx$ 0.1-0.2, where the inference sample retains enough observations for precise PPI estimation. Beyond $\rho = 0.3$, the widening confidence intervals dominate whatever gains the larger selection sample provides, as reflected in the increasing CI widths and declining power.

6 Theoretical Results

6.1 Conditional Independence

Lemma 1. *Let $\mathcal{P} = \{R_1, \ldots, R_K\}$ be any partition determined by $\mathcal{D}_{\text{sel}}$ alone. Then for each k and any $(X_i, Y_i) \in \mathcal{D}_{\text{inf}}$,*

$$\mathbb{E}[Y_i - f(X_i) \mid X_i \in R_k,\ \mathcal{D}_{\text{sel}}] = \mathbb{E}[Y_i - f(X_i) \mid X_i \in R_k] =: \delta_k. \tag{17}$$

Proof. $\mathcal{P}$ is a function of $\mathcal{D}_{\text{sel}}$, and $\mathcal{D}_{\text{sel}} \perp \mathcal{D}_{\text{inf}}$ by the random split, so conditioning on $\mathcal{D}_{\text{sel}}$ does not change the distribution of inference observations. □

The quantity $\delta_k = \mathbb{E}[Y - f(X) \mid X \in R_k]$ is the **region-specific mean residual**—the prediction bias of f within R_k. It is generally nonzero under model misspecification. The empirical rectifier $\hat{\Delta}_{R_k}$ is an unbiased estimator of δ_k, which is what allows the PPI estimator $\hat{\theta}^{\text{PP}}_{R_k} = \tilde{\theta}^{f}_{R_k} - \hat{\Delta}_{R_k}$ to be unbiased for $\theta^*_{R_k}$.

6.2 Variance Crossover Between Splitting and Randomization

From the derivation in Section 3.4 with $\eta = \rho$, the rectifier-variance terms (dropping the common $\sigma_f^2/(N\pi)$) are

$$V_{\text{split}}(\rho) = \frac{\sigma^2_{f\text{-}Y}}{(1-\rho)\, n\pi}, \qquad V_{\text{rand}}(\rho) = \frac{(1-\rho)\, \sigma^2_Y}{n\pi}.$$

Setting $V_{\text{rand}} < V_{\text{split}}$ gives $(1-\rho)^2 < \sigma^2_{f\text{-}Y}/\sigma^2_Y \approx 1 - R^2$, i.e.,

$$\rho > 1 - \sqrt{1 - R^2}. \tag{18}$$

Under the current DGP, $R^2 \approx 0.22$, which places the predicted crossover at $\rho \approx 0.12$. The variance comparison therefore predicts that the randomized approach would have lower rectifier variance than splitting for most values of ρ in this regime. Empirical validation of this prediction is left to future work (Section 7.2). When R^2 is high, sample splitting benefits more from the reduced $\sigma^2_{f\text{-}Y}$ and the crossover shifts rightward.

6.3 FDR Control for the Multi-Region Pipeline

In Section 5, we apply BH to the K region-wise PPI p-values. Conditional on $\mathcal{P}$, the test statistic for region R_k depends only on $\{(X_i, Y_i) : i \in \mathcal{D}_{\text{inf}},\ X_i \in R_k\}$ and $\{f(\tilde{X}_j) : \tilde{X}_j \in R_k\}$–non-overlapping subsets of the data across regions. The disjointness of these subsets is designed to produce a dependence structure (independence or at least PRDS) under which BH controls FDR at level $\alpha_{\text{BH}} \cdot K_0/K$, where K_0 is the number of truly null regions. We do not prove this condition formally; however, the simulation in Table 2 shows that the empirical mean FDP stays well below $\alpha_{\text{BH}} = 0.10$ across all ρ, which is consistent with the theoretical expectation.

6.4 The Optimal Split Ratio ρ^*

We define ρ^* operationally in finite simulation as the split ratio with minimum average interval width among values whose empirical coverage is within Monte Carlo uncertainty of the nominal level. In Section 4, with $B = 1000$ replications and nominal coverage 0.90, this means treating values with coverage at least $0.90 - 2SE_{MC} \approx 0.88$ as admissible. Under this criterion, $\rho = 0.1$ is optimal in the single-region experiment. In Section 5, where nominal coverage is 0.95, $\rho = 0.1$ again minimizes width while maintaining coverage above nominal. Thus, in both experiments, the optimum occurs at small ρ because interval width responds strongly to ρ whereas selection accuracy is nearly flat.

The same optimum appears in both experiments because interval width is highly sensitive to ρ, while selection quality changes little across the grid. In the single-region setting, Jaccard overlap remains roughly flat, indicating that the shallow CART selector rather than selection-sample size is the main bottleneck. Thus increasing ρ mostly sacrifices inferential precision without meaningfully improving subgroup recovery.

7 Discussion and Future Work

7.1 Main Takeaways

Our results highlight the central tradeoff in adaptive PPI when encountering group selection. The two approaches studied have different strengths. Sample splitting is conceptually simple and assumption-lean: once the partition is learned on $\mathcal{D}_{\text{sel}}$, the rectifier on $\mathcal{D}_{\text{inf}}$ is unbiased for the selected region by construction, so region-wise PPI can be applied exactly as in the classical pre-specified setting.

Randomized inference is a theoretical proposal in this report. By decomposing Y into a noisy selection view and an independent inference view, it avoids discarding labeled covariate locations, which the variance analysis in Section 3.4 suggests should matter most when the prediction model is weak or the signal-bearing subgroup is rare. Empirical validation of these analytical predictions remains an important next step.

Meanwhile, the current report makes clear that these two branches are not equally mature. The sample-splitting approach is the more fully operationalized contribution in the present draft. In Section 5, it extends naturally from single-region validity to a multi-region discovery pipeline: a tree is fit on $\mathcal{D}_{\text{sel}}$, region-wise PPI p-values are computed on $\mathcal{D}_{\text{inf}}$, BH is used for region selection, and selective interval adjustment is used for reporting. Under the simulation design considered there, this pipeline achieves low empirical false discovery proportions, high power, and near-nominal marginal coverage, while also revealing a clear selection-inference tradeoff as the split ratio ρ varies.

A second takeaway from the simulations is that adaptive validity alone does not guarantee large efficiency gains. In the multiple-region experiments, the prediction model has only modest quality, so the PPI intervals are only slightly narrower than their classical counterparts. This is not a failure of the method; rather, it reinforces one of the original paper's main messages in a new adaptive setting: the practical benefit of PPI is strongly tied to model quality. When the model explains little of the outcome variation, the rectifier variance remains large and the PPI advantage over classical inference is necessarily limited. In that sense, the adaptive setting does not overturn the logic of PPI, but sharpens it by introducing an additional allocation problem between subgroup discovery and estimation.

There is also an important robustness contrast between the two validity-restoring strategies. Sample splitting requires no special distributional structure beyond the standard asymptotic conditions already used by PPI for the rectifier. Outside the Gaussian setting, the same decomposition is still suggestive, but the exact independence argument no longer holds automatically. This makes sample splitting the more robust default, especially when the goal is to extend adaptive PPI beyond a narrow stylized model.

7.2 Future Directions

Several directions follow naturally from this work. First, cross-fitting or repeated splitting could recover some of the labeled-data efficiency lost under a single split while preserving the basic selection-inference separation that makes the sample-splitting argument work. Second, an important open problem is whether the randomized construction can be extended from single-region validity to simultaneous multi-region discovery with formal multiple-testing guarantees. That would require a careful analysis of dependence across region-wise test statistics under shared randomization. Third, it would be valuable to extend the adaptive framework beyond mean estimation to more general PPI targets such as quantiles or regression coefficients. The selective-inference issue remains the same, but the inferential machinery becomes more involved because the underlying PPI confidence sets are no longer simple one-dimensional normal intervals.

More broadly, this project suggests that the right way to think about adaptive PPI is not as a small modification of the original procedure, but as a selective-inference problem with two distinct design axes: how to separate selection from estimation, and how to spend the limited labeled information budget. Sample splitting gives a practical and robust solution that already supports multiple-region discovery. Randomized inference offers a potentially more efficient alternative whose advantages are clearest in difficult regimes, but whose theory and implementation still require further development. That tension between robustness and efficiency is, in our view, the main conceptual takeaway of the project.

8 Appendix

Section 4 Monte Carlo Simulation over the ρ-Grid

```
library(dplyr)
library(rpart)
library(partykit)
library(ggplot2)

set.seed(42)

# Global parameters
n_labeled   <- 300
n_unlabeled <- 5000
n_pretrain  <- 2000
n_eval      <- 200000
sigma_eps   <- 1
alpha       <- 0.10
B           <- 1000
rho_grid    <- seq(0.1, 0.9, by = 0.1)


# Data-generating process
generate_sample <- function(n, labeled = TRUE, sigma_eps = 1) {
  X1 <- rnorm(n)
  X2 <- rnorm(n)
  X3 <- rbinom(n, 1, 0.5)

  if (labeled) {
    Y <- 1 + 1.5 * X1 - 1.0 * X2 +
      0.8 * X3 + 1.2 * (X1 > 0 & X2 > 0) +
      rnorm(n, sd = sigma_eps)

    data.frame(X1 = X1, X2 = X2, X3 = X3, Y = Y)
  } else {
    data.frame(X1 = X1, X2 = X2, X3 = X3)
  }
}

# large independent evaluation sample
eval_df <- generate_sample(n_eval, labeled = TRUE, sigma_eps = sigma_eps)


# Monte Carlo simulation
mc_results <- data.frame()

for (b in 1:B) {

  # generate independent datasets
  pretrain_df  <- generate_sample(n_pretrain, labeled = TRUE, sigma_eps = sigma_eps)
  labeled_df   <- generate_sample(n_labeled, labeled = TRUE, sigma_eps = sigma_eps)
  unlabeled_df <- generate_sample(n_unlabeled, labeled = FALSE)

  # fit prediction model on independent pretraining data
```

```r
pred_model <- lm(Y ~ X1 + X2 + X3, data = pretrain_df)

labeled_df$fhat   <- predict(pred_model, newdata = labeled_df)
unlabeled_df$fhat <- predict(pred_model, newdata = unlabeled_df)

for (rho in rho_grid) {

  n <- nrow(labeled_df)
  n_sel <- floor(rho * n)
  sel_idx <- sample(seq_len(n), size = n_sel, replace = FALSE)

  # split labeled data into selection and inference sets
  D_sel <- labeled_df[sel_idx, , drop = FALSE]
  D_inf <- labeled_df[-sel_idx, , drop = FALSE]

  # fit shallow CART tree on residuals in the selection set
  D_sel <- D_sel %>%
    mutate(residual = Y - fhat)

  tree_fit <- rpart(
    residual ~ X1 + X2 + X3,
    data = D_sel,
    method = "anova",
    control = rpart.control(
      maxdepth = 2,
      minbucket = 10,
      cp = 0
    )
  )

  tree_party <- as.party(tree_fit)

  # assign observations to learned regions
  D_inf$region       <- predict(tree_party, newdata = D_inf,       type = "node")
  unlabeled_df$region <- predict(tree_party, newdata = unlabeled_df, type = "node")
  eval_df$region     <- predict(tree_party, newdata = eval_df,     type = "node")

  for (r in sort(unique(D_inf$region))) {
    inf_r  <- D_inf %>% filter(region == r)
    unl_r  <- unlabeled_df %>% filter(region == r)
    eval_r <- eval_df %>% filter(region == r)

    n_r <- nrow(inf_r)
    N_r <- nrow(unl_r)

    if (n_r < 2 || N_r < 2 || nrow(eval_r) < 2) next

    # PPI estimator within region r
    theta_tilde_f <- mean(unl_r$fhat)
    rectifier_hat <- mean(inf_r$Y - inf_r$fhat)
    theta_ppi_hat <- theta_tilde_f + rectifier_hat

    # plug-in variance estimate and confidence interval
    var_rect <- var(inf_r$Y - inf_r$fhat)
```

```
      var_pred <- var(unl_r$fhat)
      se_hat   <- sqrt(var_rect / n_r + var_pred / N_r)

      z_crit   <- qnorm(1 - alpha / 2)
      ci_lower <- theta_ppi_hat - z_crit * se_hat
      ci_upper <- theta_ppi_hat + z_crit * se_hat

      # approximate true parameter in region r
      theta_eval <- mean(eval_r$Y)

      # coverage indicator
      covered <- as.integer(theta_eval >= ci_lower && theta_eval <= ci_upper)

      # interval width
      width <- ci_upper - ci_lower

      # selection accuracy via Jaccard similarity
      true_region <- with(eval_df, X1 > 0 & X2 > 0)
      selected_region <- eval_df$region == r

      intersection <- sum(true_region & selected_region)
      union <- sum(true_region | selected_region)
      jaccard <- ifelse(union > 0, intersection / union, NA_real_)

      mc_results <- rbind(
        mc_results,
        data.frame(
          replication   = b,
          rho           = rho,
          region        = r,
          n_sel         = n_sel,
          n_inf_region  = n_r,
          N_unl_region  = N_r,
          theta_ppi_hat = theta_ppi_hat,
          theta_eval    = theta_eval,
          se_hat        = se_hat,
          ci_lower      = ci_lower,
          ci_upper      = ci_upper,
          covered       = covered,
          width         = width,
          jaccard       = jaccard
        )
      )
    }
  }
}

# Summaries by rho
summary_by_rho <- mc_results %>%
  group_by(rho) %>%
  summarise(
    mean_coverage = mean(covered, na.rm = TRUE),
    mean_width    = mean(width, na.rm = TRUE),
    mean_jaccard  = mean(jaccard, na.rm = TRUE),
```

```
    .groups = "drop"
  )

summary_by_rho

# empirical optimal rho*
rho_opt <- summary_by_rho %>%
  filter(mean_coverage >= 0.90) %>%
  slice_min(order_by = mean_width, n = 1)

rho_opt

# Example plots
ggplot(summary_by_rho, aes(x = rho, y = mean_coverage)) +
  geom_line() +
  geom_point() +
  geom_hline(yintercept = 0.90, linetype = "dashed") +
  labs(title = "Coverage vs rho", y = "Empirical Coverage")

ggplot(summary_by_rho, aes(x = rho, y = mean_width)) +
  geom_line() +
  geom_point() +
  labs(title = "Interval Width vs rho", y = "Average Interval Width")

ggplot(summary_by_rho, aes(x = rho, y = mean_jaccard)) +
  geom_line() +
  geom_point() +
  labs(title = "Selection Accuracy vs rho", y = "Average Jaccard Similarity")
```

Section 4 Plot Generation

```
# Coverage vs rho
p_coverage <- ggplot(summary_by_rho, aes(x = rho, y = mean_coverage)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  geom_hline(yintercept = 0.90, linetype = "dashed") +
  labs(
    title = "Empirical Coverage vs Selection Proportion",
    x = expression(rho),
    y = "Empirical Coverage"
  ) +
  theme_minimal()

# Interval width vs rho
p_width <- ggplot(summary_by_rho, aes(x = rho, y = mean_width)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    title = "Average Interval Width vs Selection Proportion",
    x = expression(rho),
    y = "Average Interval Width"
  ) +
  theme_minimal()
```

```r
# Selection accuracy vs rho
p_jaccard <- ggplot(summary_by_rho, aes(x = rho, y = mean_jaccard)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    title = "Selection Accuracy vs Selection Proportion",
    x = expression(rho),
    y = "Average Jaccard Similarity"
  ) +
  theme_minimal()

# Display plots
print(p_coverage)
print(p_width)
print(p_jaccard)
```

Section 5 Simulation Code

```
set.seed(781)
generate_data <- function(n, mu_vec, sigma_y = 1.0) {
  X1 <- rnorm(n)
  X2 <- rnorm(n)
  X3 <- rbinom(n, 1, 0.5)
  quad <- rep(4L, n)
  quad[X1 > 0 & X2 > 0]  <- 1L
  quad[X1 > 0 & X2 <= 0] <- 2L
  quad[X1 <= 0 & X2 > 0] <- 3L
  mu <- mu_vec[quad]
  Y <- mu + rnorm(n, sd = sigma_y)
  data.frame(X1 = X1, X2 = X2, X3 = X3, Y = Y, quad = quad)
}

train_predictor <- function(n_train, mu_vec, sigma_y = 1.0) {
  df <- generate_data(n_train, mu_vec, sigma_y)
  lm(Y ~ X1 + X2 + X3, data = df)
}
ppi_inference <- function(Y_inf, fhat_inf, fhat_unl, alpha = 0.05) {
  n_r <- length(Y_inf)
  N_r <- length(fhat_unl)
  if (n_r < 2 || N_r < 2) return(list(theta = NA, ci_lo = NA, ci_hi = NA, se = NA, pval = N

  theta_tilde_f <- mean(fhat_unl)
  rectifier     <- mean(fhat_inf - Y_inf)
  theta_pp      <- theta_tilde_f - rectifier
  var_rect <- var(fhat_inf - Y_inf)
  var_pred <- var(fhat_unl)
  se       <- sqrt(var_rect / n_r + var_pred / N_r)
  z_crit   <- qnorm(1 - alpha / 2)
  ci_lo    <- theta_pp - z_crit * se
  ci_hi    <- theta_pp + z_crit * se
  z_stat <- if (se > 0) theta_pp / se else 0
  pval   <- 2 * (1 - pnorm(abs(z_stat)))
  list(theta = theta_pp, ci_lo = ci_lo, ci_hi = ci_hi, se = se, pval = pval)
}

classical_inference <- function(Y_inf, alpha = 0.05) {
  n_r <- length(Y_inf)
  if (n_r < 2) return(list(theta = NA, ci_lo = NA, ci_hi = NA, se = NA, pval = NA))
  theta_hat <- mean(Y_inf)
  se        <- sd(Y_inf) / sqrt(n_r)
  z_crit    <- qnorm(1 - alpha / 2)
  ci_lo     <- theta_hat - z_crit * se
  ci_hi     <- theta_hat + z_crit * se
  z_stat    <- if (se > 0) theta_hat / se else 0
  pval      <- 2 * (1 - pnorm(abs(z_stat)))
  list(theta = theta_hat, ci_lo = ci_lo, ci_hi = ci_hi, se = se, pval = pval)
}

# One replicate of the pipeline
run_one_rep <- function(n_lab, N_unl, mu_vec, pred_model, rho,
```

```
                        alpha_bh = 0.10, alpha_ci = 0.05, sigma_y = 1.0) {
lab_df <- generate_data(n_lab, mu_vec, sigma_y)
unl_df <- generate_data(N_unl, mu_vec, sigma_y)
lab_df$fhat <- predict(pred_model, newdata = lab_df)
unl_df$fhat <- predict(pred_model, newdata = unl_df)

n_sel <- floor(rho * n_lab)   # Split labeled data
sel_idx <- sample(seq_len(n_lab), size = n_sel, replace = FALSE)
D_sel <- lab_df[sel_idx, ]
D_inf <- lab_df[-sel_idx, ]

# Fit tree on selection residuals
D_sel$resid <- D_sel$Y - D_sel$fhat

tree_fit <- rpart(
  resid ~ X1 + X2 + X3, data = D_sel, method = "anova",
  control = rpart.control(maxdepth = 2, minbucket = max(10, ceiling(0.05 * n_sel)), cp =
)
tree_party <- as.party(tree_fit)

# Assign regions
D_inf$region <- predict(tree_party, newdata = D_inf, type = "node")
unl_df$region <- predict(tree_party, newdata = unl_df, type = "node")

regions <- sort(unique(D_inf$region))
K <- length(regions)

# PPI inference per region
ppi_pvals  <- numeric(K)
cls_pvals  <- numeric(K)
true_means <- numeric(K)
ppi_widths <- numeric(K)
cls_widths <- numeric(K)
ppi_covers <- logical(K)
cls_covers <- logical(K)

for (k in seq_along(regions)) {
  r <- regions[k]
  inf_r <- D_inf[D_inf$region == r, ]
  unl_r <- unl_df[unl_df$region == r, ]

  if (nrow(inf_r) < 2 || nrow(unl_r) < 2) {
    ppi_pvals[k] <- NA; cls_pvals[k] <- NA; next
  }

  true_means[k] <- mean(mu_vec[inf_r$quad])
  ppi_res <- ppi_inference(inf_r$Y, inf_r$fhat, unl_r$fhat, alpha_ci)
  cls_res <- classical_inference(inf_r$Y, alpha_ci)
  ppi_pvals[k]  <- ppi_res$pval
  cls_pvals[k]  <- cls_res$pval
  ppi_widths[k] <- ppi_res$ci_hi - ppi_res$ci_lo
  cls_widths[k] <- cls_res$ci_hi - cls_res$ci_lo
  ppi_covers[k] <- (ppi_res$ci_lo <= true_means[k]) & (true_means[k] <= ppi_res$ci_hi)
  cls_covers[k] <- (cls_res$ci_lo <= true_means[k]) & (true_means[k] <= cls_res$ci_hi)
```

```
  }

  valid <- !is.na(ppi_pvals)
  if (sum(valid) == 0) return(NULL)

  ppi_pvals  <- ppi_pvals[valid]
  cls_pvals  <- cls_pvals[valid]
  true_means <- true_means[valid]

  # BH procedure
  ppi_rej <- p.adjust(ppi_pvals, method = "BH") <= alpha_bh
  cls_rej <- p.adjust(cls_pvals, method = "BH") <= alpha_bh
  is_null    <- abs(true_means) < 1e-8
  is_nonnull <- !is_null
  compute_metrics <- function(rej) {
    n_rej <- sum(rej)
    fdp <- if (n_rej > 0) sum(rej & is_null) / n_rej else 0
    pwr <- if (sum(is_nonnull) > 0) sum(rej & is_nonnull) / sum(is_nonnull) else NA
    c(fdp = fdp, power = pwr, n_disc = n_rej)
  }

  ppi_m <- compute_metrics(ppi_rej)
  cls_m <- compute_metrics(cls_rej)

  c(K = sum(valid),
    ppi_fdp = ppi_m["fdp"], ppi_power = ppi_m["power"], ppi_n_disc = ppi_m["n_disc"],
    ppi_coverage = mean(ppi_covers[valid]), ppi_width = mean(ppi_widths[valid]),
    class_fdp = cls_m["fdp"], class_power = cls_m["power"], class_n_disc = cls_m["n_disc"]
    class_coverage = mean(cls_covers[valid]), class_width = mean(cls_widths[valid]))
}

# full sim
run_simulation <- function(n_lab = 300, N_unl = 5000, mu_vec = c(2, 0, 1, 0),
                           pred_model, B = 500, rho_grid = seq(0.1, 0.7, by = 0.1)) {

  results <- data.frame()
  for (rho in rho_grid) {
    cat(sprintf("  rho = %.2f ...", rho))

    rep_list <- replicate(B, {
      res <- tryCatch(
        run_one_rep(n_lab, N_unl, mu_vec, pred_model, rho),
        error = function(e) NULL
      )
      if (is.null(res)) rep(NA, 11) else res
    })
    rep_mat <- t(rep_list)
    valid_rows <- !is.na(rep_mat[, 1])
    cat(sprintf(" (%d/%d valid)\n", sum(valid_rows), B))

    if (sum(valid_rows) == 0) next
    rep_mat <- rep_mat[valid_rows, , drop = FALSE]
    agg <- data.frame(
      rho = rho,
```

```
      ppi_fdp_mean = mean(rep_mat[, "ppi_fdp.fdp"]),
      ppi_power_mean = mean(rep_mat[, "ppi_power.power"], na.rm = TRUE),
      ppi_width_mean = mean(rep_mat[, "ppi_width"]),
      ppi_coverage_mean = mean(rep_mat[, "ppi_coverage"]),
      class_fdp_mean = mean(rep_mat[, "class_fdp.fdp"]),
      class_power_mean = mean(rep_mat[, "class_power.power"], na.rm = TRUE),
      class_width_mean = mean(rep_mat[, "class_width"]),
      class_coverage_mean = mean(rep_mat[, "class_coverage"])
    )
    results <- rbind(results, agg)
  }
  results
}
pred_model <- train_predictor(1000, c(2, 0, 1, 0))
results <- run_simulation(
  n_lab = 300, N_unl = 5000, mu_vec = c(2, 0, 1, 0),
  pred_model = pred_model, B = 500
)
print(results, digits = 3)
```

References

Angelopoulos, A. N., Bates, S., Fannjiang, C., Jordan, M. I., & Zrnic, T. (2023). Prediction-powered inference. *Science*, *382*, 669–674.